Explore Safely, Learn Fast: Risk-Aware Cold-Start Recommendation with LLM Contrastive Scoring

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ABSTRACT

When a new user signs up for a recommendation platform, the system knows nothing about them. This is called the *cold-start problem*, and it forces the system to make a difficult tradeoff: should it play it safe and recommend popular items, or explore new ones to learn what the user actually likes? Existing approaches like LinUCB handle this tradeoff mathematically but have a blind spot — they treat all unknown items as equally worth exploring, even if some of those items might be inappropriate or off-putting for certain users. On the other hand, using a Large Language Model (LLM) to rank items by how much a user might like them sounds promising, but LLMs tend to be overconfident when ask vague questions without any context to anchor their answers. In this project, we propose a unified scoring framework that combines three ideas to address these gaps. First, *Anchored Contrastive Scoring (ACS)* asks the LLM to compare a candidate item against thing the user has already liked or skipped, producing more reliable preference estimates. Second, *Semantic Neighborhood Disagreement (SND)* measures how uncertain we are about a user's taste in a specific content area by checking how inconsistently they have rated similar items in the past. Third, a *Risk Penalty* $R(a)$ discourage the system from exploring items that could be harmful or alienating before we know anything about the user. Together, these three components are combined into a single score that balances what the user probably likes, what we still need to learn about them, and what is safe to show them.

I. INTRODUCTION

Every recommendation system faces a fundamental chicken-and-egg problem with new users: to make good recommendations you need to know what someone likes, but to learn what they like you have to recommend things first. This is the *cold-start problem*, and it shows up not just for brand new users but also whenever a user switches context — for example, a teenager browsing a platform for the first time, or an existing user exploring a completely new content category.

The standard approach to this problem comes from a field called *contextual bandits*. The idea is simple: the system treats each recommendation as a gamble, and over time it learns which gambles pay off for which kinds of users. The most well-known algorithm in this family is **LinUCB** [1], which works by maintaining a model of what each user probably likes and then adding an exploration bonus to items it is still uncertain about — the logic being that the more uncertain you are about something, the more you can learn by trying it. LinUCB works reasonably well in practice, but it has two problems that motivate this project.

The first problem is that LinUCB is *content-blind*. Its exploration bonus is purely geometric — it depends on where an item sits in a mathematical feature space, not on what the item actually is. This means it might eagerly recommend a disturbing or niche piece of content simply because it has not been tried before, with no consideration of whether that content is appropriate for a user whose preferences are still completely unknown.

The second problem is that LinUCB's measures of uncertainty is not the right kind of uncertainty for our problem. It track how confident the system is about a single global set of reward parameters, which gets smaller as any user provides data. What we actually want to track is how uncertain we are about *this specific user's* taste in *this specific content area* — a much more local and meaningful notion of uncertainty that should shrink as the user rate more items similar to the candidate.

A natural alternative is to use a Large Language Model (LLM) to score items based on their content and what the user has liked before. LLMs have broad world knowledge and can reason about semantic similarity, making them attractive for zero-shot cold-start settings. But there is a catch: when you ask an LLM a vague questions like "will this user like this movie?" without giving it anything concrete to compare against, its answers tend to be overconfident and poorly calibrated. The model has no anchor.

This project proposes a scoring framework that aims to fix all three of these issues at once, combining LLM-based preference estimation with a better uncertainty measure and an explicit safety constraint. Our three main ideas are:

- **Anchored Contrastive Scoring (ACS):** Instead of asking the LLM "will the user like this?" in isolation, we ask it to compare the candidate item against things the user has already liked and things they have already skipped. This anchoring gives the LLM a concrete reference point and produces much more calibrated probability estimates.
- **Semantic Neighborhood Disagreement (SND):** Rather than measuring global parameter uncertainty like LinUCB, SND looks at how inconsistently the user has rated items that are semantically similar to the candidate. High disagreement among those neighbors means we genuinely do not know how this user feels about this content area, so it is worth exploring. As the user rates more similar items, this uncertainty naturally collapses toward zero.
- **Risk Penalty $R(a)$:** Some content is inherently polarizing — it tends to be either loved or strongly disliked, and recommending it to a user we know nothing about is a gamble with asymmetric downside. The risk penalty is a pre-computed score for each item that suppresses explo-

ration through such content until the system has enough information about the user to make that call responsibly.

These three components are combined into a single scoring formula:

$$Q(a,h_t) = P_{\text{ACS}}(a,h_t) + \alpha \cdot \text{SND}(a,t) - \lambda \cdot R(a) \qquad (1)$$

where P_{ACS} captures how much the user probably likes item a, SND captures how much we still need to learn about their taste in that area, and $R(a)$ penalizes items that are risky to explore blindly. The weights α and λ control the balance between exploration and safety.

The rest of this report is organized as follows. Section 3 reviews the existing work that our approach builds on. Section 4 describes each component of the framework in detail. Section 8 discusses our immediate implementation plan and what we plan to explore further.

II. LITERATURE REVIEW

Can large language models understand preferences in personalized recommendation?: A core problem in evaluating personalized recommendation is that standard regression metrics such as MAE and RMSE can be fooled by two trivially simple tricks — predicting a user's average historical rating, or predicting an item's average rating across all users — neither of which uses any personal preference signal, yet both match or beat sophisticated LLM-based methods [3]. To address this, Tan et al. introduce **PerRecBench**, which restructures evaluation as a grouped ranking task: users who all purchased the same item within a short time window are grouped together, and models must rank those users by how much each preferred the item relative to their own baseline (actual rating minus personal average). This design eliminates both user rating bias and item quality , leaving only genuine personalization signal. Benchmarking 19 LLMs reveals that even the best model achieves a Kendall's τ of only 0.18, pairwise and listwise prompting substantially outperforms pointwise, including user profiles improve performance by 28%, and PerRecBench scores shows near-zero correlation with MAE/RMSE — confirming that traditional metrics measure the wrong thing entirely.

Personalized LLM decoding via contrasting personal preference: On the generation side of personalization, Bu et al. propose **COPE** (Contrasting Personal Preference), a decoding-time framework that steers token selection toward individual user preferences without requiring an external reward model [4] at all. The key insight is that after fine-tuning a lightweight LoRA adapter per user, two models exists simultaneously: a personalized model and the generic base model. At every generation step, COPE computes an implicit user reward as the log-likelihood ratio between these two models and selects the token that maximizes this ratio, filtered by a threshold inorder to prevent irrelevant tokens.To sharpen this signal, COPE additionally applies DPO training using synthetically generated negatives; candidate responses with the lowest implicit reward are selected as hard negatives, explicitly pushing the personalized model to widen the reward gap between user-written and generic outputs. Evaluated on five open-ended generation tasks, COPE achieves a 10.57% average ROUGE-L improvement over the task-adapted baseline and generalizes consistently across LLaMA, Gemma, and Qwen model families.

Contextual bandit approach-based recommendation system for personalized web-based services: Pilani et al. model personalized recommendation as a **contextual multi-armed bandit** problem, where the system must balance recommending items that are already known to work well with exploring new items to learn more about a user's preferences [5]. Each item is treated as an arm, and its expected reward for a user is estimated using a linear function of user and item features. The paper compare three algorithms: **LinUCB**, **Hybrid-LinUCB**, and **CoLin**, where CoLin additionally shares information between similar users to improve learning in cold-start settings. Experiments on Yahoo! Front Page, LastFM, and MovieLens20M show that CoLin achieves the lowest regret due to this collaborative information sharing.

This work highlight the importance of exploration when learning user preferences. However, the exploration signal in contextual bandits is derived from model uncertainty in the feature space. In contrast, our approach introduces **Semantic Neighborhood Disagreement (SND)**, which measures uncertainty directly from the user's past reactions to semantically similar items, allowing explorations to be guided by disagreement in the user's own preference history rather than purely geometric uncertainty.

Knowledge Graph Convolutional Networks for Recommender Systems: Wang et al. propose Knowledge Graph Convolutional Networks (KGCN) to improve recommendation under sparse interaction data [6]. The model uses a knowledge graph where items are connected to related entities such as genres, actors, and directors. By aggregating information from these neighboring entities, KGCN learns richer item representations that capture semantic relationships beyond user interactions.

This idea is related to our approach in that both methods use semantic neighborhoods of items. However, while KGCN uses these neighborhoods to improve item representation learning, our method uses semantic neighborhoods in the embedding space to measure disagreement in a user's past reactions and guide exploration during recommendation.

Catching dark signals in algorithms: Unveiling audiovisual and thematic markers of unsafe content recommended for children and teenagers: In this paper, they demonstrate empirically that unsafe content directed at children and teenagers carries reproducible, structured audiovisual and thematic signatures — like aggressive sound design, dark chrome palettes, and scary narrative structures — that are detectable through analysis of low-level signal features [7]. The main relation for recommendation system is that psychological risk is not too subjective to quantify: it can be derived as a reproducible scalar from a catalog item's raw features, independent of any user interactions. The population studied — childrens and teenager with limited established preference history — is kind of analogous to a cold-start user, meaning the mentioned failure modes of unrestrained

exploration are directly related to the idea of a static content safety constraints that is observed during the time before any reliable preferences vector about the user accumulated.

Re-ranking Using Large Language Models for Mitigating Exposure to Harmful Content on Social Media Platforms: This paper demonstrate that LLMs can assess and rank content by exposure harm potential in zero-shot and few-shot settings, without needing any labelled training data [8]. Their core finding is that LLM-based pairwise harm comparison outperform traditional classifier-based moderation significantly, establishing that prompted LLMs carry sufficient world knowledge to produce reliable harm judgments from item descriptions alone. This directly validates the mechanism behind our offline risk weight assignment: prompting the LLM once per catalog item to elicit a harm scalar $R(a) \in [0,1]$ is something we draw inspiration from the authors, and the zero-shot setting they study is also the setting we intend to work in.

III. EXISTING BASELINE MODELS AND RELATED WORK

A. Classical Contextual Bandits

The contextual bandit framework formalizes sequential recommendation as a regret minimization problem. At each timestep t, the agent observes context (item features) and selects an arm (item) to maximize the expected sum of rewards.

LinUCB [1] defines the scoring function:

$$Q_{\text{LinUCB}}(a) = \hat{\boldsymbol{\theta}}_t^\top \mathbf{x}_a + \alpha\sqrt{\mathbf{x}_a^\top \mathbf{A}_t^{-1}\mathbf{x}_a} \tag{2}$$

where $\mathbf{A}_t = \sum_i \mathbf{x}_{a_i}\mathbf{x}_{a_i}^\top + \lambda\mathbf{I}$ is the regularized covariance matrix. The second term provides an upper confidence bound on the reward estimate. LinUCB achieves $\mathscr{O}(\sqrt{T\log T})$ cumulative regret in theory.

B. LLM-Augmented Bandit Methods

A recent line of work augments classical bandits with LLM-derived representations or synthetic data. Alamdari et al. [2] propose initializing contextual bandits by prompting an LLM to generate a prior dataset of approximate human preferences, giving the bandit a head start before any real interactions occur. This reduces early regret but still relies on classical UCB uncertainty at inference time, and the LLM-generated prior reflects population-level knowledge rather than anything specific to the current user.

IV. PROPOSED METHODOLOGY

A. Problem Formulation

We consider a sequential recommendation problem. At each timestep t, a user has accumulated an interaction history:

$$h_t = \{(a_1, r_1), (a_2, r_2), \ldots, (a_{t-1}, r_{t-1})\} \tag{3}$$

where a_i is the item shown at step i and $r_i \in \{0,1\}$ is the binary reward (1 = engaged, 0 = skipped). We define the positive and negative sub-histories:

$$h_t^+ = \{a_i \in h_t : r_i = 1\} \tag{4}$$

$$h_t^- = \{a_i \in h_t : r_i = 0\} \tag{5}$$

At $t = 1$, $h_1 = \emptyset$, the *cold-start moment*. The system selects the next item as:

$$a_t^* = \operatorname*{argmax}_{a \in \mathscr{C}_t} Q(a, h_t) \tag{6}$$

where $\mathscr{C}_t$ is the candidate pool at time t.

Table 1. Proposed Methodology Scoring Components (Online Phase)

Term	Measures	LLM?
P_{ACS}	Contrastive user-item fit	Yes (batched)
$\text{SND}(a,t)$	User-local epistemic uncertainty	No
$R(a)$	Stranger-risk of the item	No (cached)

B. Unified Objective Function

The proposed methodology defines a three-component scoring function:

$$\boxed{Q(a,h_t) = P_{\text{ACS}}(a,h_t) + \alpha \cdot \text{SND}(a,t) - \lambda \cdot R(a)} \tag{7}$$

where $\alpha > 0$ control exploration aggressiveness and $\lambda > 0$ controls the strength of the safety penalty. Table 1 summarizes the three components.

C. Offline Preprocessing

The following preprocessing is performed *once per catalog* and cached for online use.

1) Semantic Embeddings

A frozen LLM or sentence encoder maps every item to a dense vector:

$$f_{\text{LLM}} : a \mapsto \mathbf{e}_a \in \mathbb{R}^d \tag{8}$$

These vectors are indexed into a FAISS structure supporting approximate nearest neighbor (ANN) retrieval in $\mathscr{O}(\log|\mathscr{I}|)$.

2) Risk Weight Assignment

The LLM is prompted once per item with the query: *"On a scale from 0 to 1, how risky is it to recommend this item to a complete stranger with no established taste profile? Consider: how niche it is, how much prior context it assumes, how polarizing its content is."* This produces a static risk lookup table $R : \mathscr{I} \to [0,1]$, where $R(a) \approx 0$ denotes a safe item and $R(a) \approx 1$ denotes a highly risky item for strangers.

D. Online Pipeline

1) ANN Candidate Retrieval

The user's current preference centroid is computed from positive interactions:

$$\bar{\mathbf{e}}_t = \frac{\sum_i r_i \cdot \mathbf{e}_{a_i}}{\sum_i r_i + \varepsilon} \tag{9}$$

where $\varepsilon = 10^{-8}$ prevents division by zero. At cold-start ($h_t = \emptyset$), $\bar{\mathbf{e}}_t \approx \mathbf{0}$ and the ANN returns globally popular items—a psycho-

logically safe default. The candidate set is:

$$\mathscr{C}_t = \text{ANN}(\bar{\mathbf{e}}_t, k = M) \setminus \{a_i : (a_i, \cdot) \in h_t\} \tag{10}$$

with $M \approx 50$ in practice.

E. Term 1: Anchored Contrastive Scoring (ACS)

1) Motivation

Standard LLM rankers extract preference probabilities via:

$$P_{\text{abs}}(\text{YES} \mid a, h_t) = \text{softmax}(\log p(\text{YES}), \log p(\text{NO}))_{\text{YES}} \tag{11}$$

This yield miscalibrated estimates because the LLM must infer preferences from an unanchored item description. LLMs are known to exhibit overconfidence on cold absolute binary judgments.

2) ACS Mechanism

For each candidate $a \in \mathscr{C}_t$, we retrieve the most semantically proximate items from each sub-history:

$$x^+(a) = \underset{a_i \in h_t^+}{\text{argmin}} \|\mathbf{e}_a - \mathbf{e}_{a_i}\|_2 \tag{12}$$

$$x^-(a) = \underset{a_i \in h_t^-}{\text{argmin}} \|\mathbf{e}_a - \mathbf{e}_{a_i}\|_2 \tag{13}$$

The LLM is then queried contrastively:

> *"A user engaged with [x^+] but skipped [x^-]. Is [a] more similar to what they engaged with or what they skipped? Answer only: [ENGAGED] or [SKIPPED]."*

The ACS score is extracted from raw logprobs:

$$P_{\text{ACS}}(a, h_t) = \frac{e^{\log p(\text{ENGAGED})}}{e^{\log p(\text{ENGAGED})} + e^{\log p(\text{SKIPPED})}} \tag{14}$$

This is a binary softmax over the two forced-choice tokens, yielding a calibrated probability without requiring auxiliary calibration datasets. When $|h_t^+| = 0$ or $|h_t^-| = 0$, the system falls back to P_{abs} until both anchors are available.

F. Term 2: Semantic Neighborhood Disagreement (SND)

1) Why Entropy Fails

Shannon entropy $H(a,t)$ is maximized at $P = 0.5$ and measures the LLM's population-level prior uncertainty about an item. Crucially, it does not decrease as the *current user* provides evidence: a polarizing item will yield $P \approx 0.5$ for every user, regardless of history depth. This is aleatoric, item-specific uncertainty—not the epistemic, user-specific uncertainty required for active learning.

2) SND Definition

For candidate $a \in \mathscr{C}_t$, we retrieve the k nearest already-rated items from the user's history:

$$\mathscr{N}_k(a,t) = \text{top-k}_{a_i \in h_t} \text{ by } \|\mathbf{e}_a - \mathbf{e}_{a_i}\|_2 \tag{15}$$

Define the neighbor positive and negative counts:

$$n^+(a,t) = |\{a_i \in \mathscr{N}_k(a,t) : r_i = 1\}| \tag{16}$$

$$n^-(a,t) = |\{a_i \in \mathscr{N}_k(a,t) : r_i = 0\}| \tag{17}$$

Then:

$$\text{SND}(a,t) = \frac{\min(n^+, n^-)}{\max(n^+, n^-) + 1} \tag{18}$$

SND is zero when the neighborhood is unanimous (all liked or all skipped) and approache 0.5 when the neighborhood is maximally split.

3) ε-Floor for Cold-Start

At early timesteps, $|\mathscr{N}_k(a,t)| < 2$ for most candidates, causing SND ≈ 0 everywhere. To ensure occasional probing of unexplored semantic regions, we apply a small floor:

$$\text{SND}_\varepsilon(a,t) = \text{SND}(a,t) + \varepsilon \cdot \mathbf{1}[|\mathscr{N}_k(a,t)| < 2] \tag{19}$$

with $\varepsilon = 0.05$.

4) Convergence Property

Theorem 1. *For any user with consistent preferences near item a:*

$$\lim_{t \to \infty} \mathbb{E}[SND(a,t)] = 0 \tag{20}$$

Proof Sketch. If the user consistently engages with items near a, then for $t \geq k$ all k rated neighbors of a have $r_i = 1$, giving $n^- = 0$ and $\text{SND}(a,t) = 0$ from (18). Symmetrically for consistent skips. In the case of genuine preference inconsistency, SND stabilizes at a non-zero value reflecting true aleatoric variance or decreases as the user's preferences evolve. Critically, $\text{SND}(a,t)$ is non-increasing in expectation: each interaction either preserves or reduces neighborhood disagreement. □ □

This is in direct contrast to Shannon entropy, which is non-decreasing with interaction count and thus unsuitable as an epistemic exploration signal.

G. Term 3: Risk Penalty $R(a)$

The term $-\lambda \cdot R(a)$ creates an *asymmetric exploration policy*. For safe items ($R(a) \approx 0$), the penalty is negligible and the system may explore freely; for risky items ($R(a) \approx 1$), the penalty is large and can only be overcome by a very high P_{ACS} score. The key safety invariant of the proposed methodology is:

> *A high-SND, high-$R(a)$ item—uncertain AND risky—has its exploration bonus cancelled by the risk penalty. Exploration proceeds only through safe items.*

This is an important departure from constrained-MDP safe exploration method, which learn safety constraints online and thus provide no guarantee at cold-start.

V. ARCHITECTURE

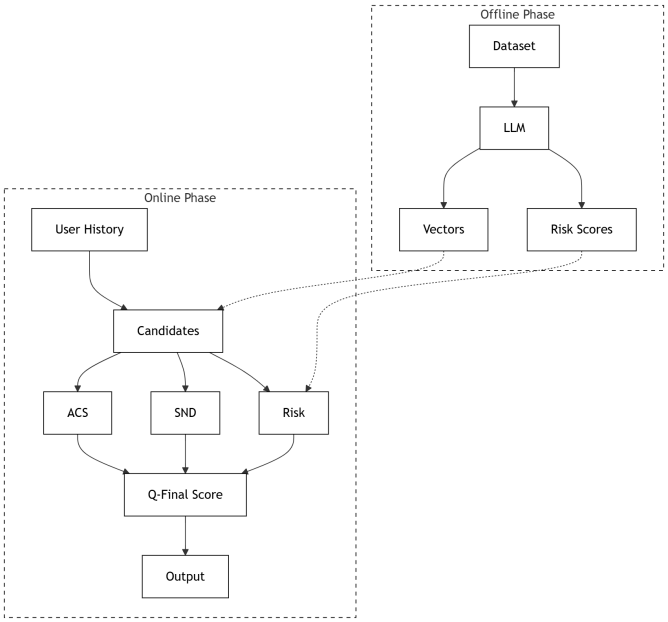

Figure 1. Overview of the proposed framework. The offline phase produces semantic embeddings and risk scores for all catalog items. At inference time, the online pipeline retrieves candidates via ANN search, scores each via ACS, SND, and $R(a)$, and selects the item with the highest $Q(a, h_t)$.

VI. APPLICATIONS

We evaluate our framework on two domains that are well-suited to the specific challenges our framework addresses: cold-start personalization and irrelevant exposure prevention, which are **movie recommendations** and **scientific paper recommendation**.

VII. PROPOSED DATASETS

A. MovieLens

MovieLens contain explicit user ratings for movie on a 1–5 star scale, along with timestamps and genre metadata. We plan to use the **MovieLens-1M** variant. It is well-suited to our framework because movie preferences are highly personal, cold-start failures are immediately visible, and the catalog naturally spans a wide range of exposure harm risk levels.

B. Semantic Scholar Open Data

The Semantic Scholar Open Data is a large collections of academic papers with titles, abstracts, citation links, and field-of-study tags. It complements MovieLens because exposure harms here is context-sensitive rather than content-specific, papers abstracts provides rich natural language for semantic reasonings, and research topics form well-defined cluster in embedding space.

VIII. CONCLUSION

We have presented our proposed methodology for cold-start sequential recommendation. By decomposing the scoring function into three orthogonal components—ACS for personalized exploitation, SND for epistemic exploration, and $R(a)$ for risk penalty—the proposed methodology addresses the three principal failure modes of LinUCB while retaining the bandit framing.

A. Implementation and Future Work

Moving forward, our immediate focus is translating this theoretical framework into a functional pipeline by conducting offline preprocessing of semantic embeddings and static risk scores $R(a)$, followed by implementing the online retrieval system to dynamically compute ACS and SND. Once operational, we plan to benchmark the system's regret minimization and safe exploration against classical baselines like LinUCB using the MovieLens and Semantic Scholar datasets. Also, in future we will try to make risk scaling dynamic to safely relax the risk penalty as the user's preference history matures.

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